

Norepinephrine bitartrate

(Levophed)

I. Classification – catecholamine

II. Action(s)

- Produces peripheral vasoconstriction by direct effects on the α -adrenergic receptors
- Produces inotropic stimulation and coronary artery dilation by direct effects on the β_1 -adrenergic receptors

III. Indication(s)

- Shock / severe hypotension

IV. Contraindications

- Known Hypersensitivity
- Hypovolemia
- Hypertension
- Profound hypoxia or hypercarbia – risk of inducing cardiac arrhythmias
- Sulfite sensitivity

V. Precautions

- Hypoxia, hypercapnia and acidosis, may reduce the effectiveness and/or increase the incidence of adverse effects of norepinephrine and must be identified and corrected prior to or concurrently with administration of the drug.
- Avoid extravasation (infiltration) of norepinephrine. Tissue necrosis may occur.
- Avoid infusion into leg veins, especially in geriatric patients or those with occlusive vascular diseases, arteriosclerosis, diabetes mellitus, or Buerger's disease.
- Use with caution in patients with sulfite sensitivity, especially if sulfites induce asthma
- Use caution when administering to patients with hyperthyroid. Norepinephrine can cause severe hypertension with headache, photophobia, stabbing chest pain, pallor, sweating and vomiting

VI. Drug interactions

- MAOI's
- Tricyclic antidepressants – may cause severe, prolonged hypertension
- Antihistamines – may cause severe, prolonged hypertension

VII. Adult Dosing and Administration

- For hypotension not responsive to fluid resuscitation (1-2 liters):
 - Mix 8 mg (8 mL) in 500 mL NS, infuse via pump
 - Typically, Start 4 mcg/min IV / IO, titrate 1 mcg/min - 12 mcg/min, maintain SBP >90
 - Range 0.05 – 2 mcg/kg/min

VIII. Pediatric Dosing and Administration

- For hypotension not responsive to fluid resuscitation (1-2 liters)
 - Mix 8 mg (8 mL) in 500 mL NS, infuse via pump
 - 0.05 mcg/kg/min IV / IO, range 0.05 – 2 mcg/kg/min to maintain SBP >90 or age based normal

IX. Onset

- Rapid

X. Duration

- Short; pressor action stops within 1-2 minutes after the infusion is discontinued

XI. Possible Adverse Effects

- Hypertension
- Headache
- Anxiety
- Photophobia
- Cardiac Arrhythmias
- Extravasation necrosis
- Reflex bradycardia

XII. Other Notes

- Closely monitor the infusion flow rate.
- Adequate intravascular volume replacement is strongly recommended before, during, and upon withdrawal of therapy. Inadequate intravascular volume can result in decreased perfusion of vital organs, including the heart, liver, and kidney.
- Check BP every 2 minutes from the time the norepinephrine infusion is started until the desired effect is achieved, then every 5 minutes while the drug is being infused.
- Use a large bore IV catheter (18 ga or greater) in a large peripheral vein.
- Discontinuing therapy: Slow infusion rate gradually and avoid abrupt withdrawal: observe patient carefully so that therapy may be resumed if the BP falls too rapidly.
- Can increase glycogenolysis and inhibit insulin release in the pancreas, resulting in hyperglycemia.